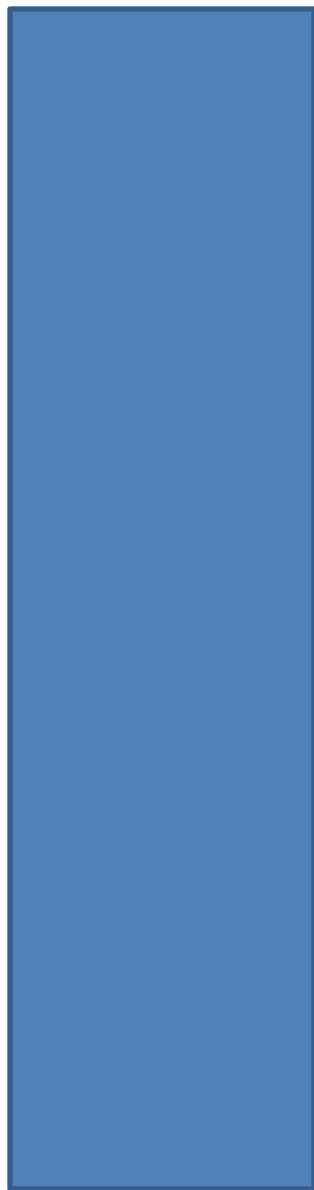
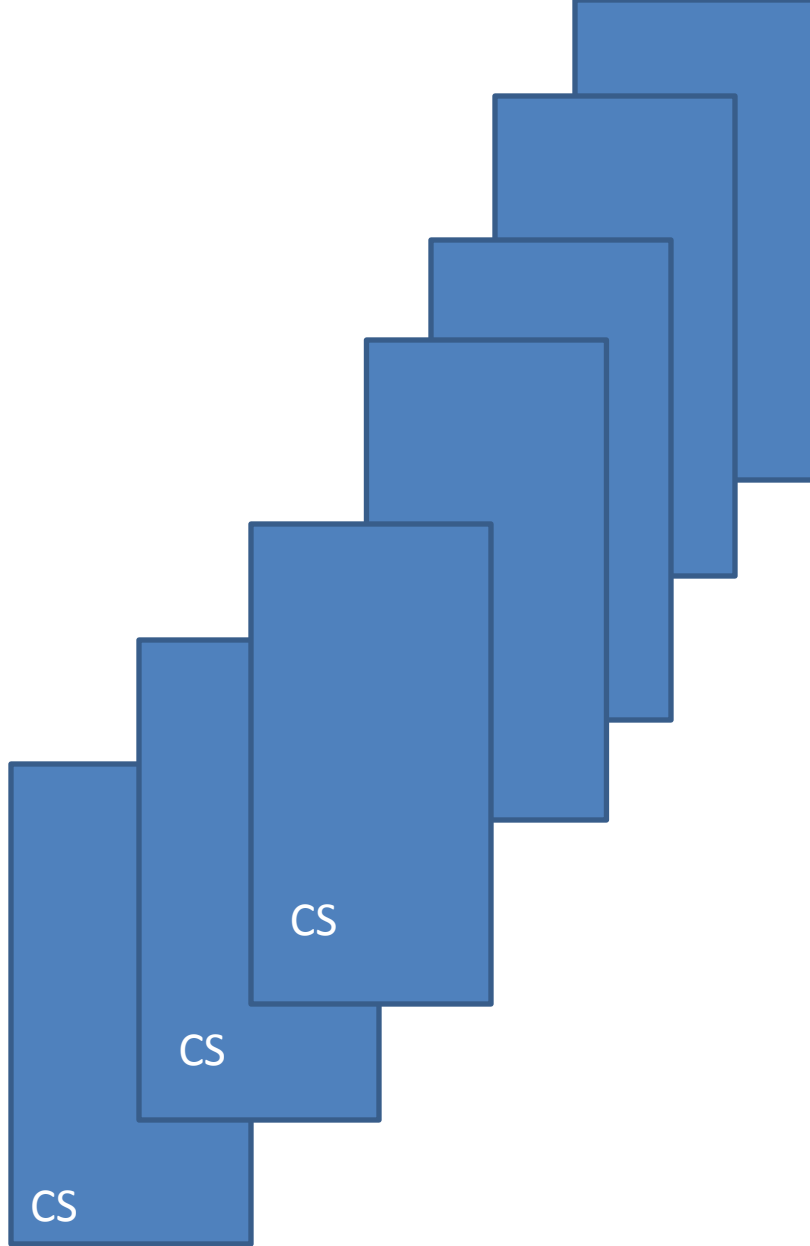


Section 12

Sheet 12 solution

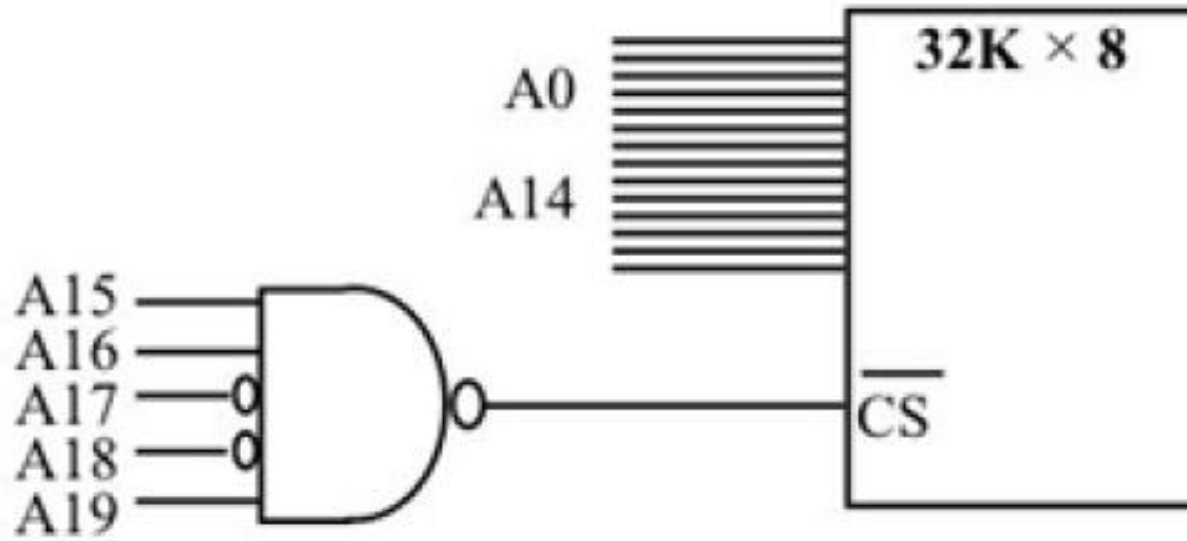


One Chip



Multiple chips

1.



$A_{19} A_{18} A_{17} A_{16} A_{15} A_{14} A_{13} A_{12} A_{11} A_{10} A_9 A_8 A_7 A_6 A_5 A_4 A_3 A_2 A_1 A_0$

100110000000000000000000=98000

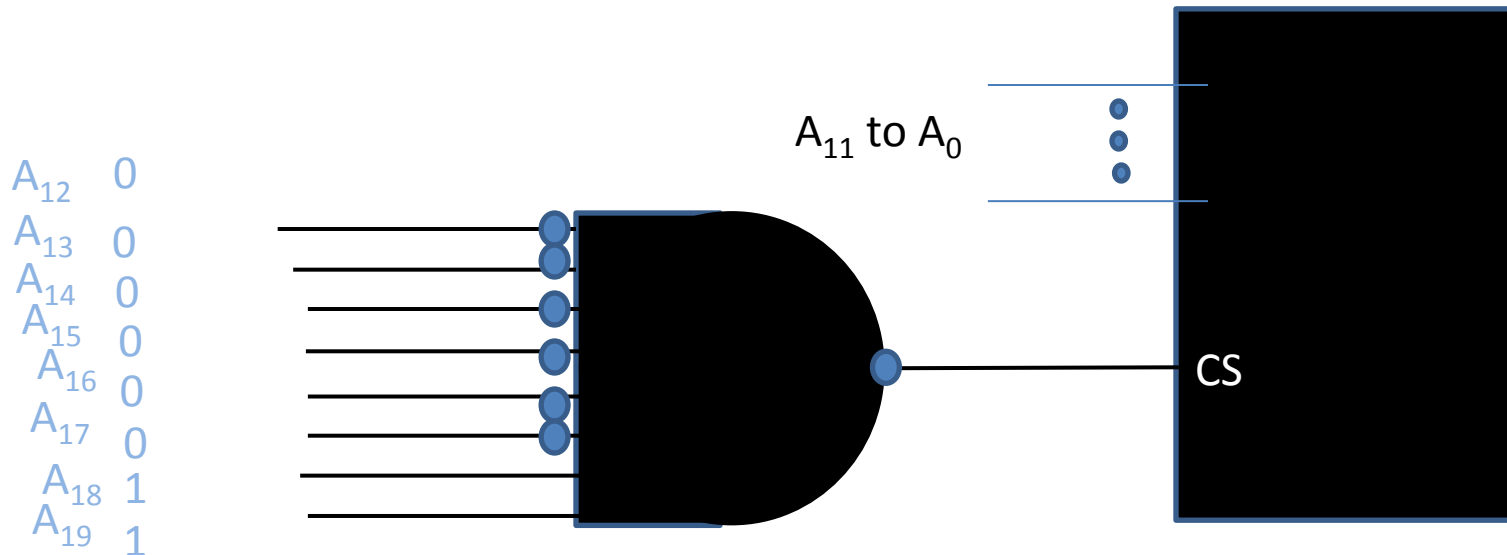
100111111111111111111111=9FFFF

2.

$A_{19} A_{18} A_{17} A_{16} A_{15} A_{14} A_{13} A_{12} A_{11} A_{10} A_9 A_8 A_7 A_6 A_5 A_4 A_3 A_2 A_1 A_0$

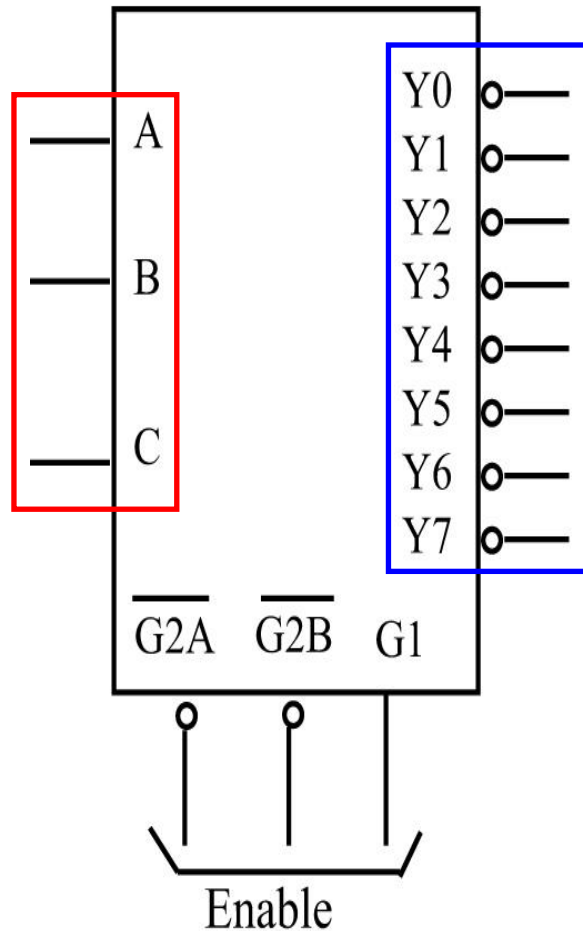
110000000000000000000000=C0000

11000000111111111111=C0FFF



Three inputs:
A, B & C,
 generate
 eight
active-low
 outputs:
Y0–Y7.

Block Diagram

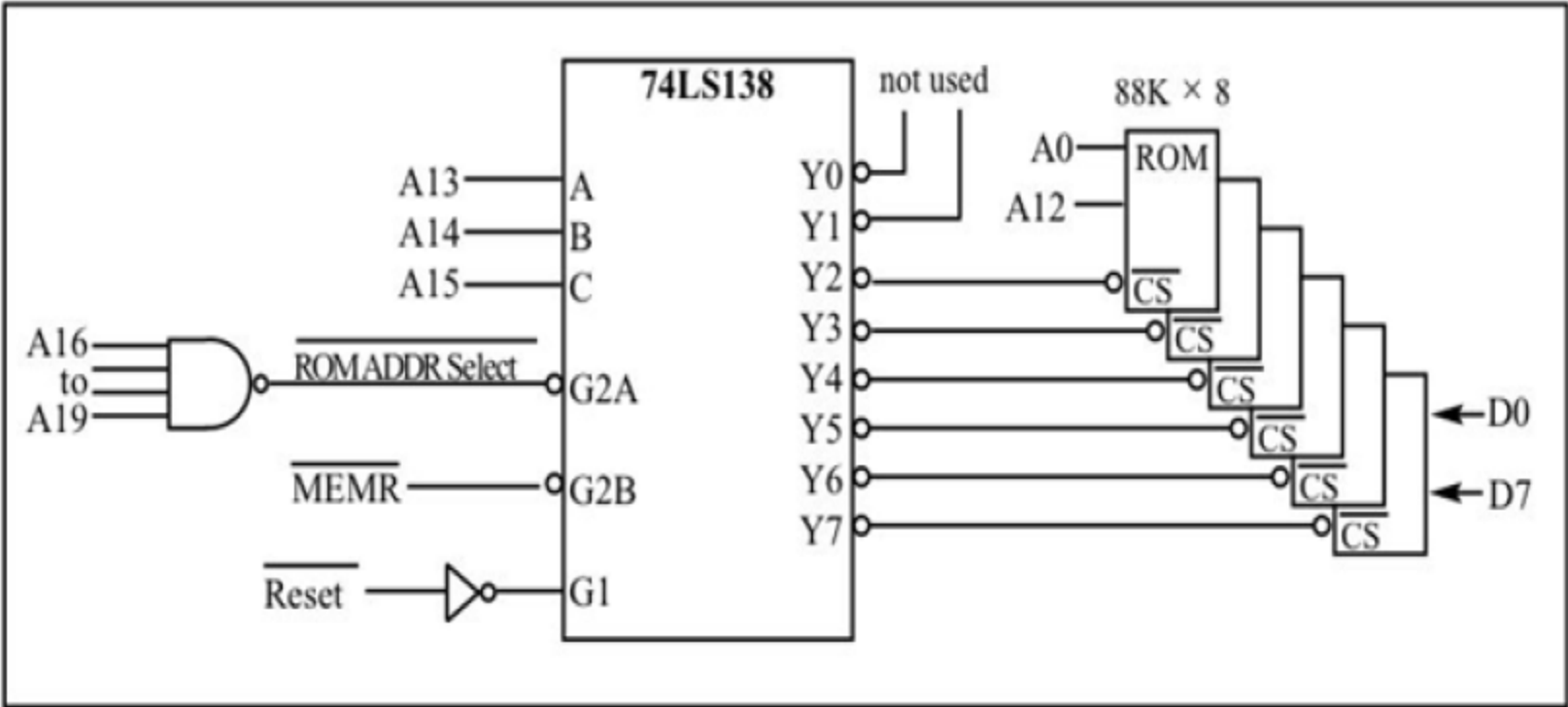


Function Table

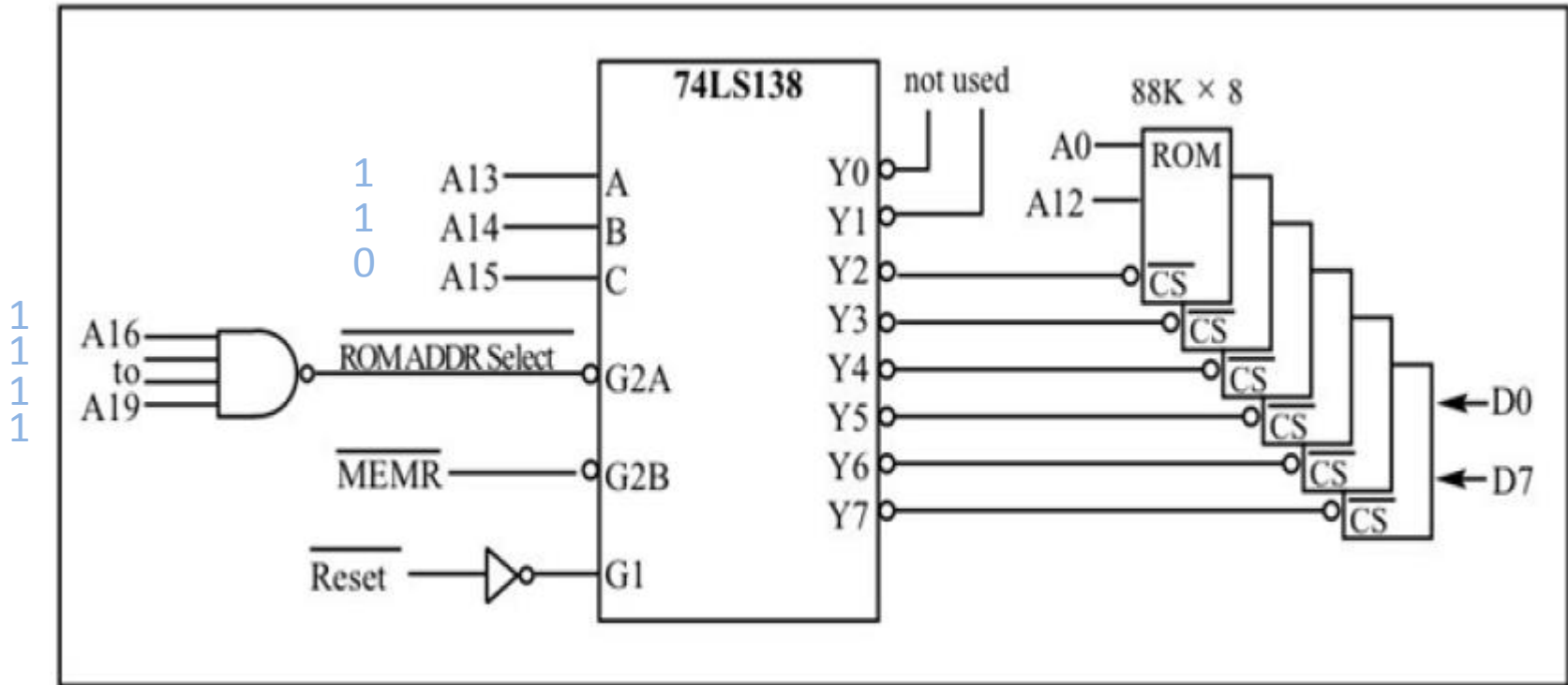
Inputs									
Enable	Select	Outputs							
G1G2	C B A	Y0	Y1	Y2	Y3	Y4	Y5	Y6	Y7
X H	X X X	H	H	H	H	H	H	H	H
L X	X X X	H	H	H	H	H	H	H	H
H L	L L L	L	H	H	H	H	H	H	H
H L	L L H	H	L	H	H	H	H	H	H
H L	L H L	H	H	L	H	H	H	H	H
H L	L H H	H	H	H	L	H	H	H	H
H L	H L L	H	H	H	H	L	H	H	H
H L	H L H	H	H	H	H	H	L	H	H
H L	H H L	H	H	H	H	H	H	L	H
H L	H H H	H	H	H	H	H	H	H	L

Fig. 10-11 74LS138 Decoder

3. Find the address range for Y0, Y3, and Y6 of the 74LS138 for the following design. This is the ROM interfacing with the 8088 CPU in the original PC.



Y3

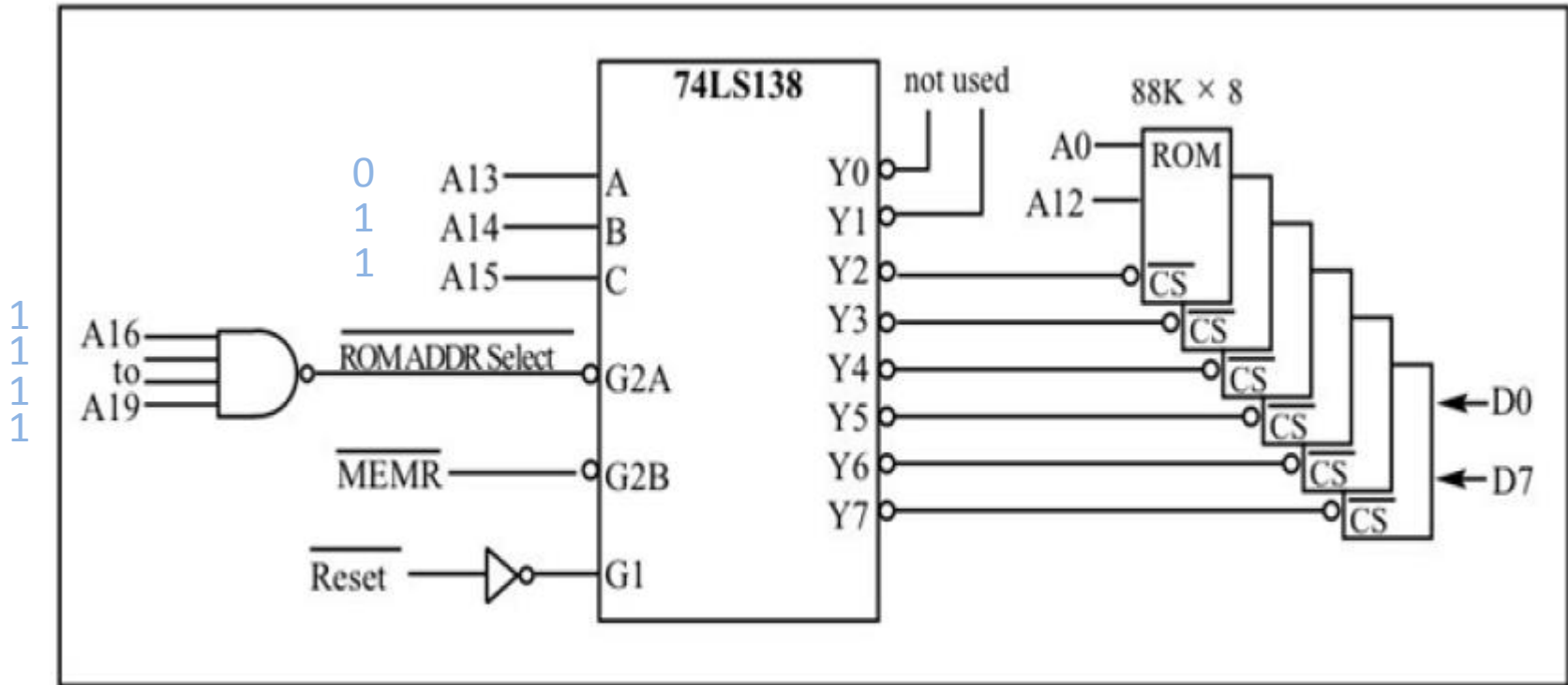


A₁₉ A₁₈ A₁₇ A₁₆ A₁₅ A₁₄ A₁₃ A₁₂ A₁₁ A₁₀ A₉ A₈ A₇ A₆ A₅ A₄ A₃ A₂ A₁ A₀

111101100000000000000000=F6000

111101111111111111111111=F7FFF

Y6



A₁₉ A₁₈ A₁₇ A₁₆ A₁₅ A₁₄ A₁₃ A₁₂ A₁₁ A₁₀ A₉ A₈ A₇ A₆ A₅ A₄ A₃ A₂ A₁ A₀

111111000000000000000000=FC000

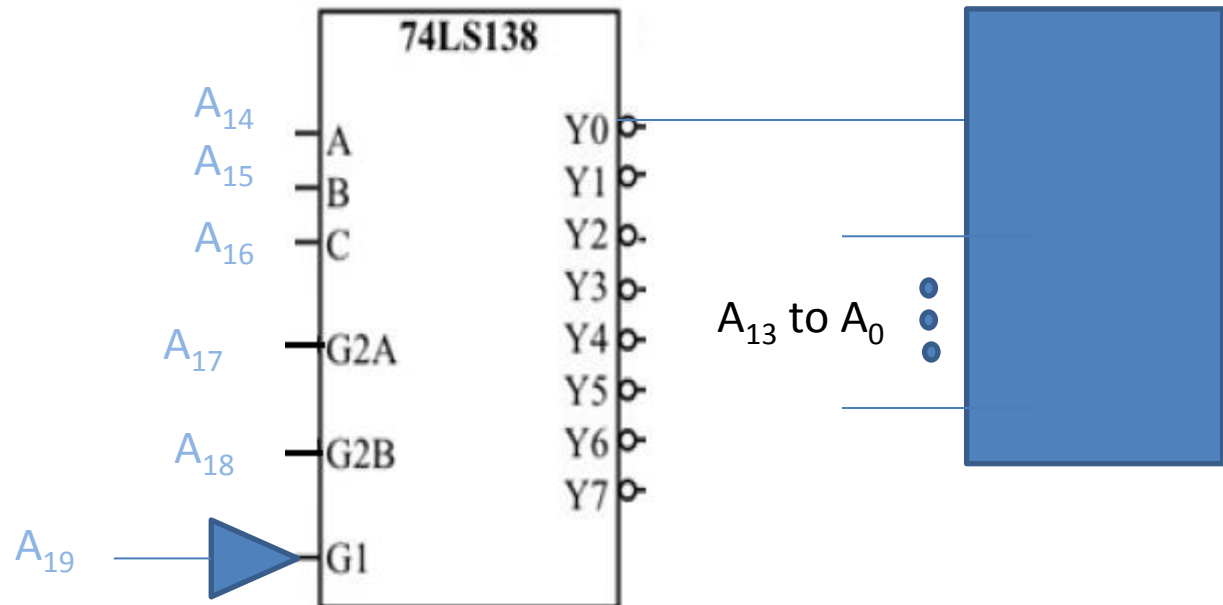
111111011111111111111111=FDFFF

4.

A_{19} A_{18} A_{17} A_{16} A_{15} A_{14} A_{13} A_{12} A_{11} A_{10} A_9 A_8 A_7 A_6 A_5 A_4 A_3 A_2 A_1 A_0

00000000000000000000000000000000=000000

00000001111111111111111111111111=03FFF

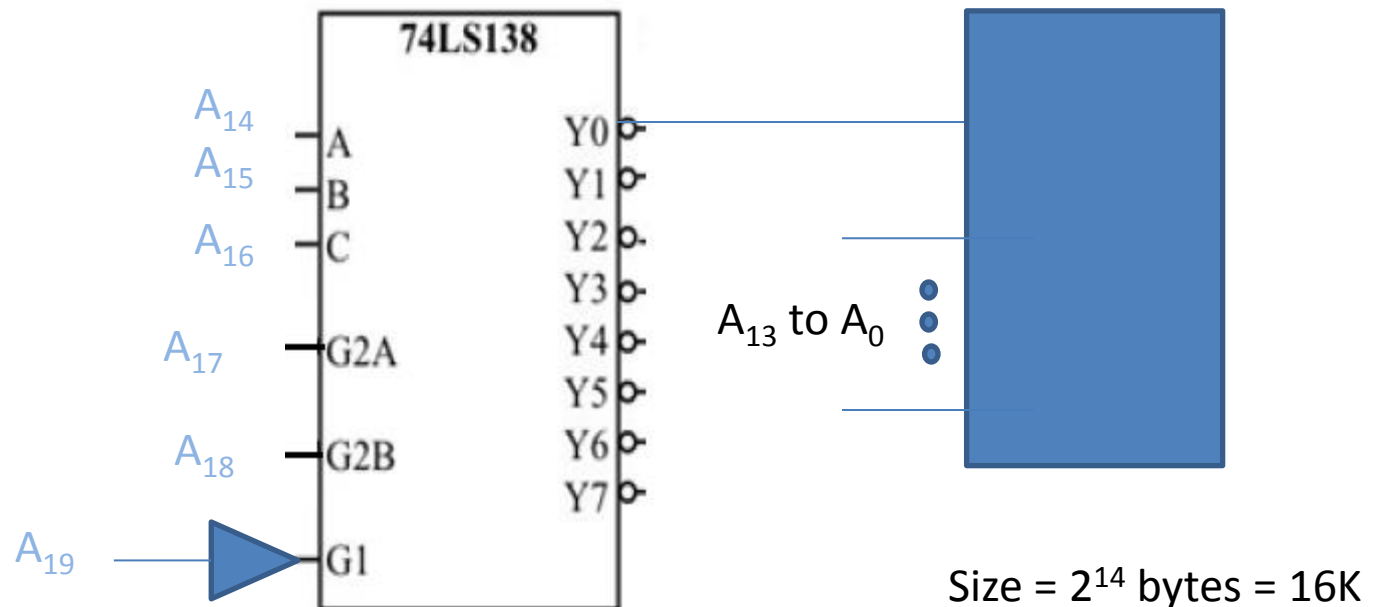


4.

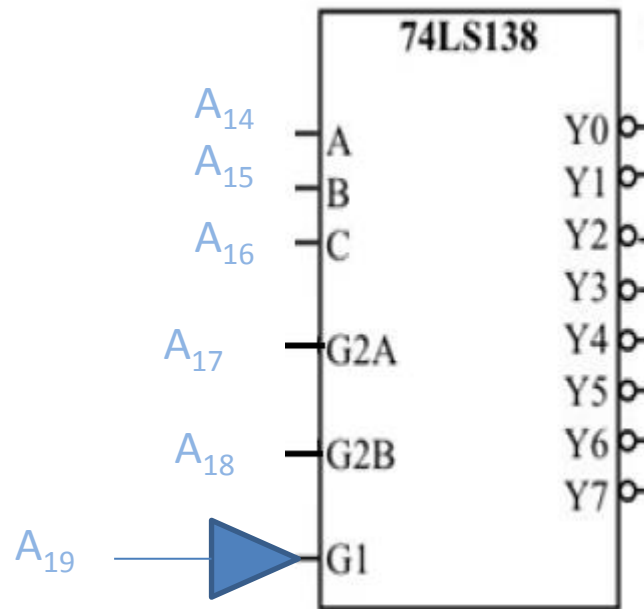
C B A
 A_{19} A_{18} A_{17} A_{16} A_{15} A_{14} A_{13} A_{12} A_{11} A_{10} A_9 A_8 A_7 A_6 A_5 A_4 A_3 A_2 A_1 A_0

00000000000000000000000000000000=000000

00000001111111111111111111111111=03FFF



5. Range of Y3



C B A
 $A_{19} A_{18} A_{17} A_{16} A_{15} A_{14} A_{13} A_{12} A_{11} A_{10} A_9 A_8 A_7 A_6 A_5 A_4 A_3 A_2 A_1 A_0$

000011000000000000000000=0C000

000011111111111111111111=0FFFF

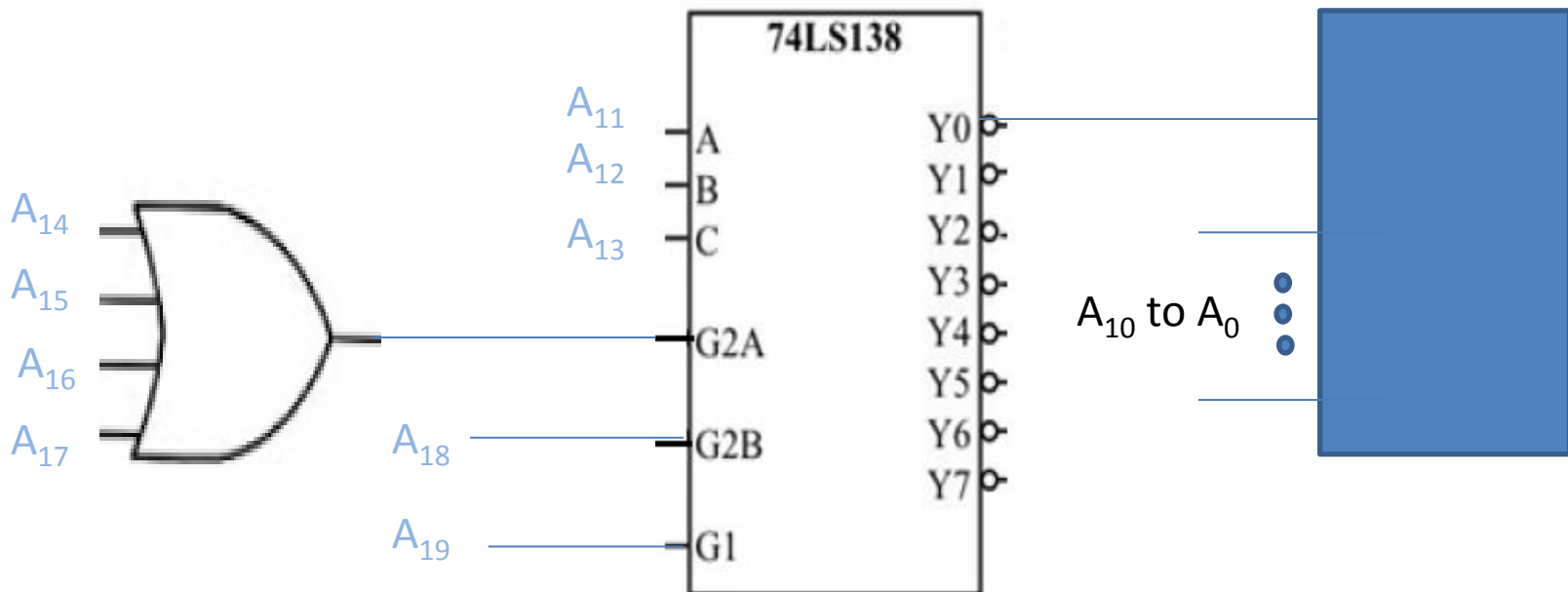
6.

C B A

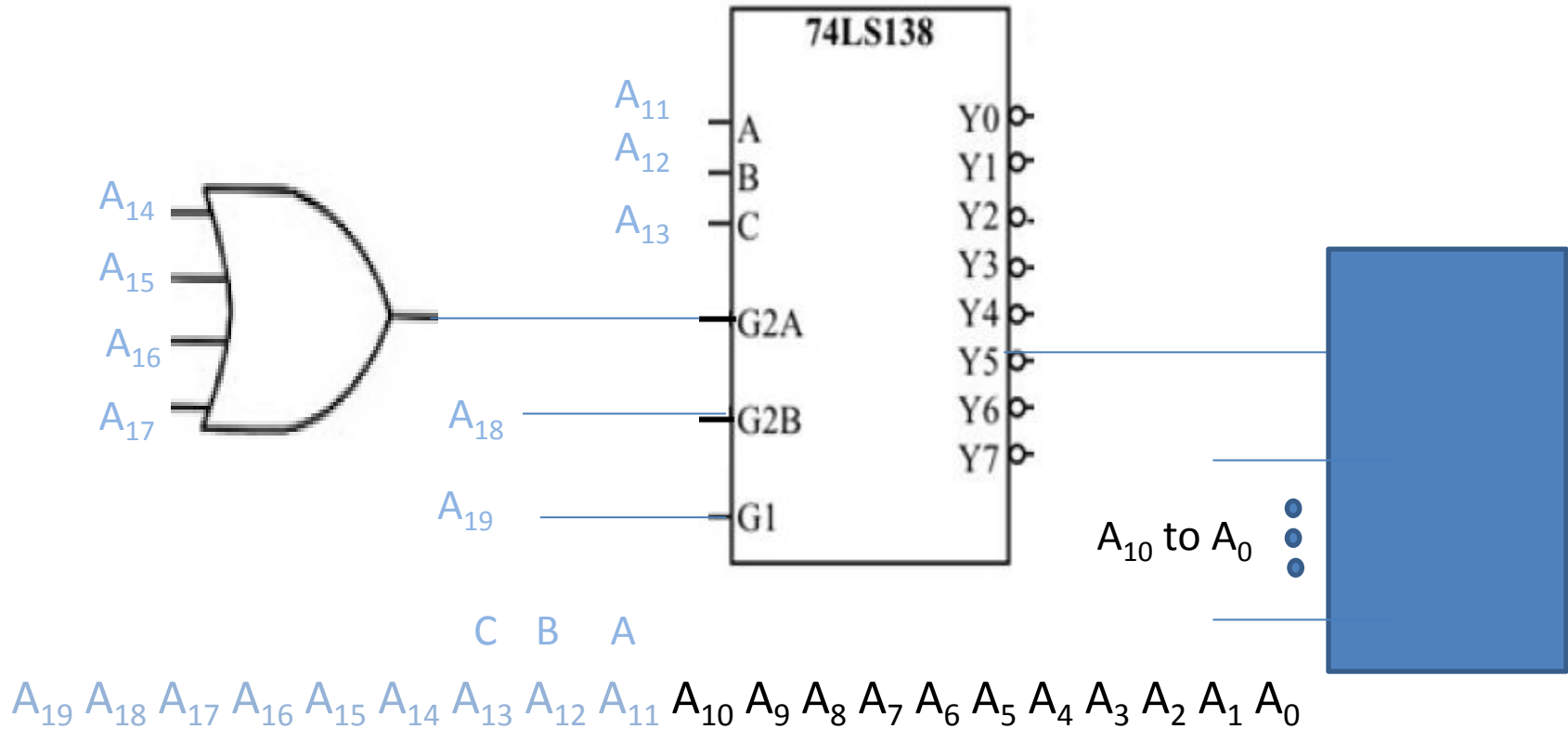
$A_{19} A_{18} A_{17} A_{16} A_{15} A_{14} A_{13} A_{12} A_{11} A_{10} A_9 A_8 A_7 A_6 A_5 A_4 A_3 A_2 A_1 A_0$

10000000000000000000000000000000=80000

100000000001111111111111111111=807FF



7. Range of Y5



100000101000000000000000=82800

100000101111111111111111=82FFF

8. Low, Low

9. CPLD